1 Probability Fundamentals

1.1 Basic Probability Concepts

Sample Space and Events

The sample space (Ω) is the set of all possible outcomes of an experiment. An event is a subset of the sample space.

Example 1.1.1 (Coin Flips).

- Sample space: $\Omega = \{H, T\}$
- Event (getting heads): $E = \{H\}$
- Probability: $P(H) = \frac{1}{2}$

Example 1.1.2 (Rolling a Die).

- Sample space: $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Event (rolling even): $E = \{2, 4, 6\}$
- Probability: $P(E) = \frac{3}{6} = \frac{1}{2}$

Probability Axioms

1. **Non-negativity:** $P(E) \geq 0$ for any event E
2. **Normalization:** $P(\Omega) = 1$
3. **Additivity:** For mutually exclusive events, $P(A \cup B) = P(A) + P(B)$

1.2 Conditional Probability

The conditional probability of A given B is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{1}$$

Example 1.2.1 (Card Drawing). Suppose we draw a card from a standard deck:

- $P(\text{King}) = \frac{4}{52} = \frac{1}{13}$
- $P(\text{Heart}) = \frac{13}{52} = \frac{1}{4}$
- $P(\text{King} \cap \text{Heart}) = \frac{1}{52}$
- $P(\text{King}|\text{Heart}) = \frac{P(\text{King} \cap \text{Heart})}{P(\text{Heart})} = \frac{1/52}{1/4} = \frac{1}{13}$

Example 1.2.2 (Medical Testing). A disease affects 1% of the population. A test has 95% sensitivity (true positive rate) and 90% specificity (true negative rate). What's the probability someone has the disease given a positive test?

- $P(\text{Disease}) = 0.01$
- $P(\text{Positive}|\text{Disease}) = 0.95$
- $P(\text{Positive}|\text{No Disease}) = 0.10$
- Using Bayes' theorem (see below), $P(\text{Disease}|\text{Positive}) \approx 0.087$ or 8.7%

1.3 Bayes' Theorem

Bayes' theorem is fundamental to machine learning, especially in Bayesian methods and classification:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)} \tag{2}$$

Example 1.3.1 (Email Spam Classification)**.** Given that 30% of emails are spam, and the word "free" appears in 60% of spam emails but only 5% of legitimate emails, what's the probability an email with "free" is spam?

- $P(\text{Spam}) = 0.30$, $P(\text{Legitimate}) = 0.70$
- $P(\text{“free”}|\text{Spam}) = 0.60$
- $P(\text{“free”}|\text{Legitimate}) = 0.05$
- $P(\text{“free”}) = P(\text{“free”}|\text{Spam}) \times P(\text{Spam}) + P(\text{“free”}|\text{Legitimate}) \times P(\text{Legitimate})$
- $P(\text{“free”}) = 0.60 \times 0.30 + 0.05 \times 0.70 = 0.215$
- $P(\text{Spam}|\text{“free”}) = \frac{0.60 \times 0.30}{0.215} \approx 0.837$ or 83.7%

1.4 Random Variables

A random variable is a function that maps outcomes from the sample space to real numbers.

Example 1.4.1 (Discrete Random Variable)**.** Let X be the number of heads in 3 coin flips:

- X can take values: $\{0, 1, 2, 3\}$
- $P(X = 0) = \frac{1}{8}$ (TTT)
- $P(X = 1) = \frac{3}{8}$ (HTT, THT, TTH)
- $P(X = 2) = \frac{3}{8}$ (HHT, HTH, THH)
- $P(X = 3) = \frac{1}{8}$ (HHH)

Example 1.4.2 (Continuous Random Variable)**.** Height of adult males follows a normal distribution with mean $\mu = 70$ inches and standard deviation $\sigma = 3$ inches. The probability density function is:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{3}$$

1.5 Expectation and Variance

Expectation (Mean)

For discrete random variables:

$$E[X] = \sum_x x \cdot P(X = x) \tag{4}$$

For continuous random variables:

$$E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx \tag{5}$$

Example 1.5.1 (Expected Value of Die Roll)**.**

$$E[X] = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6} = 3.5 \tag{6}$$

Variance

The variance measures the spread of a distribution:

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2 \tag{7}$$

Example 1.5.2 (Variance of Die Roll)**.**

$$E[X^2] = 1^2 \times \frac{1}{6} + 2^2 \times \frac{1}{6} + \cdots + 6^2 \times \frac{1}{6} = \frac{91}{6} \tag{8}$$

$$\text{Var}(X) = \frac{91}{6} - (3.5)^2 = \frac{91}{6} - 12.25 \approx 2.92 \tag{9}$$

1.6 Common Probability Distributions

Bernoulli Distribution

Models a single trial with two outcomes (success/failure). Parameter: p (probability of success).

Example 1.6.1 (Click Prediction)**.** Whether a user clicks an ad can be modeled as Bernoulli($p = 0.05$).

- $P(\text{click}) = 0.05$
- $P(\text{no click}) = 0.95$

Binomial Distribution

Number of successes in n independent Bernoulli trials. Parameters: n (trials), p (success probability).

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \tag{10}$$

Example 1.6.2 (Quality Control)**.** Out of 100 manufactured items, if each has a 2% defect rate, the number of defects $X \sim \text{Binomial}(n = 100, p = 0.02)$.

- Expected defects: $E[X] = np = 2$

Normal (Gaussian) Distribution

The most important continuous distribution in ML. Parameters: μ (mean), σ^2 (variance).

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \tag{11}$$

Example 1.6.3 (Test Scores)**.** Test scores follow $N(\mu = 75, \sigma^2 = 100)$.

- About 68% of scores fall within $\mu \pm \sigma$ (65-85)
- About 95% within $\mu \pm 2\sigma$ (55-95)

1.7 Joint and Marginal Distributions

Joint distribution describes the probability of multiple random variables occurring together. Marginal distribution is obtained by summing/integrating over other variables.

Example 1.7.1 (Student Performance)**.** Consider study time (Low/High) and exam result (Pass/Fail).

Joint distribution $P(\text{Study}, \text{Result})$:

- $P(\text{Low}, \text{Fail}) = 0.35$
- $P(\text{Low}, \text{Pass}) = 0.15$
- $P(\text{High}, \text{Fail}) = 0.05$
- $P(\text{High}, \text{Pass}) = 0.45$

Marginal distribution $P(\text{Study})$:

- $P(\text{Low}) = 0.35 + 0.15 = 0.50$
- $P(\text{High}) = 0.05 + 0.45 = 0.50$

1.8 Independence and Covariance

Independence

Two events A and B are independent if:

$$P(A \cap B) = P(A) \times P(B) \tag{12}$$

Example 1.8.1 (Independent Events)**.** Flipping two fair coins:

- $P(\text{First} = H) = \frac{1}{2}$
- $P(\text{Second} = H) = \frac{1}{2}$
- $P(\text{Both} = H) = \frac{1}{4} = \frac{1}{2} \times \frac{1}{2}$

The events are independent.

Covariance and Correlation

Covariance measures how two variables vary together:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] \tag{13}$$

Correlation normalizes covariance:

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \rho \leq 1 \tag{14}$$

Example 1.8.2 (Height and Weight)**.** Height and weight are positively correlated ($\rho \approx 0.7$). Taller people tend to weigh more, though not perfectly.

- If $\rho = 0$: uncorrelated
- If $\rho = 1$: perfectly positively correlated
- If $\rho = -1$: perfectly negatively correlated

2 Linear Algebra Fundamentals

2.1 Vectors

A vector is an ordered array of numbers. In ML, vectors represent features, data points, or model parameters.

Example 2.1.1 (Feature Vector)**.** A house might be represented as:

$$\mathbf{x} = \begin{bmatrix} 2000 \\ 3 \\ 2 \\ 1995 \end{bmatrix} \tag{15}$$

(square feet, bedrooms, bathrooms, year built)

Vector Operations

Example 2.1.2 (Vector Addition)**.** If position$_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and displacement $= \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, then:

$$\text{position}_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix} \tag{16}$$

Example 2.1.3 (Scalar Multiplication)**.** If velocity $= \begin{bmatrix} 5 \\ 12 \end{bmatrix}$ m/s, after 3 seconds:

$$\text{displacement} = 3 \times \begin{bmatrix} 5 \\ 12 \end{bmatrix} = \begin{bmatrix} 15 \\ 36 \end{bmatrix} \text{ meters} \tag{17}$$

2.2 Dot Product and Norms

Dot Product

For vectors $\mathbf{a}$ and $\mathbf{b}$, the dot product is:

$$\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i \tag{18}$$

Example 2.2.1 (Computing Similarity)**.** User preferences $\mathbf{a} = \begin{bmatrix} 5 & 3 & 0 & 4 \end{bmatrix}^T$ and movie features $\mathbf{b} = \begin{bmatrix} 4 & 4 & 2 & 3 \end{bmatrix}^T$:

$$\text{Similarity} = \mathbf{a} \cdot \mathbf{b} = 5 \times 4 + 3 \times 4 + 0 \times 2 + 4 \times 3 = 44 \tag{19}$$

Vector Norms

L2 norm (Euclidean):

$$\|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2} \tag{20}$$

L1 norm (Manhattan):

$$\|\mathbf{x}\|_1 = \sum_i |x_i| \tag{21}$$

Example 2.2.2 (Distance Calculation)**.** Points $\mathbf{p}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{p}_2 = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$:

$$\text{L2 distance} = \|\mathbf{p}_2 - \mathbf{p}_1\|_2 = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = 5 \tag{22}$$

$$\text{L1 distance} = \|\mathbf{p}_2 - \mathbf{p}_1\|_1 = |4-1| + |6-2| = 3 + 4 = 7 \tag{23}$$

2.3 Matrices

A matrix is a 2D array of numbers. In ML, matrices represent datasets, transformations, or weights.

Example 2.3.1 (Data Matrix)**.** Dataset of 3 houses with 2 features (size, price):

$$X = \begin{bmatrix} 1500 & 300000 \\ 2000 & 400000 \\ 1800 & 350000 \end{bmatrix} \tag{24}$$

Each row is a data point; each column is a feature.

2.4 Matrix Operations

Matrix Multiplication

For A (m×n) and B (n×p), the product $C = AB$ has dimensions (m×p) where:

$$C_{ij} = \sum_k A_{ik} B_{kj} \tag{25}$$

Example 2.4.1 (Linear Transformation)**.** Transform point $\mathbf{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ by rotation matrix A:

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \tag{26}$$

For $\theta = 90$:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad A\mathbf{x} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix} \tag{27}$$

Transpose

The transpose A^T flips rows and columns: $(A^T)_{ij} = A_{ji}$

Example 2.4.2 (Transpose Operation)**.**

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \tag{28}$$

2.5 Matrix Inverse and Determinant

Matrix Inverse

For a square matrix A, if A^{-1} exists, then:

$$AA^{-1} = A^{-1}A = I \tag{29}$$

where I is the identity matrix.

Example 2.5.1 (2×2 Matrix Inverse)**.**

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \tag{30}$$

Verification:

$$AA^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \tag{31}$$

Determinant

For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$\det(A) = ad - bc \tag{32}$$

Example 2.5.2 (Determinant Calculation)**.**

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}, \quad \det(A) = 3 \times 4 - 2 \times 1 = 12 - 2 = 10 \tag{33}$$

2.6 Eigenvalues and Eigenvectors

An eigenvector $\mathbf{v}$ of matrix A satisfies:

$$A\mathbf{v} = \lambda\mathbf{v} \tag{34}$$

where λ is the eigenvalue.

Example 2.6.1 (Finding Eigenvectors)**.** For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$:

- Eigenvalue $\lambda_1 = 3$, eigenvector $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- Eigenvalue $\lambda_2 = 1$, eigenvector $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Verification:

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \checkmark \tag{35}$$

2.7 Systems of Linear Equations

Linear systems $A\mathbf{x} = \mathbf{b}$ appear throughout ML, especially in linear regression.

Example 2.7.1 (Solving a Linear System)**.** Solve: $2x + y = 5$ and $x + y = 3$

Matrix form:

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix} \tag{36}$$

Solution:

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \tag{37}$$

So $x = 2$, $y = 1$.

2.8 Matrix Decompositions

Eigendecomposition

For a square matrix A with n linearly independent eigenvectors, the eigendecomposition is:

$$A = Q\Lambda Q^{-1} \tag{38}$$

where Q is a matrix whose columns are the eigenvectors of A, and Λ is a diagonal matrix whose diagonal elements are the corresponding eigenvalues.

For symmetric matrices, Q is orthogonal (i.e., $Q^{-1} = Q^T$), so:

$$A = Q\Lambda Q^T \tag{39}$$

Example 2.8.1 (Eigendecomposition of a 2×2 Matrix)**.** For the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ from Example 2.11:

- Eigenvalues: $\lambda_1 = 3$, $\lambda_2 = 1$
- Eigenvectors: $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

  Normalizing the eigenvectors:

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \tag{40}$$

  The eigendecomposition is:

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \tag{41}$$

Singular Value Decomposition (SVD)

Any matrix A can be decomposed as:

$$A = U\Sigma V^T \tag{42}$$

where U and V are orthogonal matrices and Σ is diagonal.

Example 2.8.2 (SVD of a 2×2 Matrix)**.** Consider the matrix:

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 0 \end{bmatrix} \tag{43}$$

First, compute $A^T A$:

$$A^T A = \begin{bmatrix} 3 & 2 \\ 2 & 0 \end{bmatrix}^T \begin{bmatrix} 3 & 2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 13 & 6 \\ 6 & 4 \end{bmatrix} \tag{44}$$

The eigenvalues of $A^T A$ are $\lambda_1 = 16$ and $\lambda_2 = 1$, giving singular values:

$$\sigma_1 = \sqrt{16} = 4, \quad \sigma_2 = \sqrt{1} = 1 \tag{45}$$

The corresponding normalized eigenvectors form V:

$$V = \begin{bmatrix} \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} \tag{46}$$

Computing AA^T and its eigenvectors gives:

$$U = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \end{bmatrix} \tag{47}$$

The complete SVD is:

$$A = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} \tag{48}$$